

NovaByte Solutions — Task/Case Study 3

SQL Investigation: Detailed Findings Report

Marketing & Support — Every query, every result, what it means

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Career 247 — Business Analytics with GenAI

About This Report

This document is the narrative companion to NovaByte_Task3_SQL_Script.sql — all 8 queries (26 through 33), run against the cleaned novabyte_capstone database. For each one: the business question being answered, the approach taken, and the finding in plain English. The full runnable SQL lives in the .sql file itself; this document is for reading the story, not re-running the code.

Queries 26-28 cover Marketing. Queries 29-31 cover Support. Queries 32-33 are cross-functional, joining Sales, Support, and Clients together.

Marketing Queries (26-28)

Query 26

Spend, MQLs, Conversions, and CAC by Channel

Business question: which marketing channels are we spending the most on, and which ones actually deliver customers efficiently?

Approach: Groups every campaign by Channel and rolls up total spend, total MQLs, total conversions, and the average Customer Acquisition Cost across that channel's campaigns.

	Channel	Total_Spend	Total_MQLs	Total_Conversions	Avg_CAC
▶	Google Ads	18925265	47612	4344	14956.56
	Webinar	16969524	30679	2882	20658.26
	LinkedIn	16088043	44736	3618	29827.79
	Content Marketing	15220453	37386	2927	73986.12
	SEO	15211008	20681	1813	28655.07
	Social Media	15098107	28981	2258	17446.00
	Trade Show	11266045	15328	1605	73863.86
	Email	10472070	32937	2653	95675.02

Finding: Google Ads has the largest spend **and** the lowest CAC of any channel — it's both the biggest and the most cost-efficient. Email is the opposite extreme: the smallest total spend of any channel but by far the worst CAC — over 6x more expensive per acquired customer than Google Ads.

Query 27

Campaign ROI Ranking — Individual and by Type

Business question: which specific campaigns, and which types of campaigns, deliver the best return on the money spent?

Approach: Part 1 ranks every individual campaign by $\text{Conversions} \div \text{Budget}$ using `RANK() OVER`. Part 2 rolls the same ratio up to the Campaign_Type level (Brand Awareness, Lead Generation, etc.), averaging across all campaigns of that type, with a campaign count alongside so a "best" type based on just one or two campaigns can be spotted as less reliable.

	Type_Rank	Campaign_Type	Avg_ROI	Campaign_Count
▶	1	Brand Awareness	0.000553	34
	2	Lead Generation	0.000274	34
	3	Event Promotion	0.000244	25
	4	Retargeting	0.000197	30
	5	Product Launch	0.000193	27

Finding: Brand Awareness campaigns have the highest average ROI — roughly double the next-best type, Lead Generation. Since both top types have the same sample size (34 campaigns each), this isn't one lucky outlier dragging the average up; Brand Awareness genuinely, consistently outperforms.

Query 28

Marketing Channel → Closed Deal Value (Joined to Sales)

Business question: which marketing channels don't just generate leads, but generate deals that actually close?

Approach: Joins *Marketing_Campaigns* to *Sales_Deals* via *Lead_Source*, first mapping the two tables' different naming conventions onto each other (*Email Campaign*→*Email*, *LinkedIn Campaign*→*LinkedIn*, *Organic Search*→*SEO*), while deliberately leaving *Cold Outreach*, *Referral*, and *Unknown* as their own categories, since they have no tracked-campaign equivalent — itself a finding worth reporting, not an error to paper over.

	Mapped_Channel	Total_Deals	Deals_Won	Closed_Deal_Value
►	Trade Show	314	95	28721940
	Email	329	88	22352433
	Google Ads	287	74	21075544
	Cold Outreach	315	79	20687459
	SEO	279	77	20296501
	LinkedIn	299	62	17577554
	Referral	302	71	17329770
	Webinar	298	61	16780909
	Unknown	77	17	3558389

Finding: Trade Show tops closed-deal value even though it isn't the spend or ROI leader in Queries 26/27 — offline, in-person leads are converting well despite a smaller marketing push. Cold Outreach, a channel with zero tracked marketing spend at all, outperforms LinkedIn and Referral in raw closed value — suggesting Sales is generating real revenue outside the Marketing-tracked funnel entirely.

Support Queries (29-31)

Query 29

SLA Compliance Rate by Priority and by Product

Business question: is the support team actually meeting its response-time commitments, and if not, where is it breaking down?

Approach: *Compliance is calculated against tickets with a known outcome only (SLA_Met = Yes or No) — tickets marked Unknown are shown as their own column so they stay visible, rather than being silently dropped from the denominator or counted as a miss.*

By Priority

	Priority	Met	Missed	Unknown_Outcome	SLA_Compliance_Rate
►	Critical	138	261	9	34.59
	High	363	633	27	36.45
	Medium	570	902	43	38.72
	Low	364	571	37	38.93
	Unspecified	32	47	3	40.51

By Product

	Product	Met	Missed	Unknown_Outcome	SLA_Compliance_Rate
►	CloudShift Platform	223	434	24	33.94
	AnalyticsEdge	240	427	21	35.98
	AI Insights Module	235	414	15	36.21
	DevOps Accelerator	240	401	22	37.44
	SecureNet Suite	252	372	20	40.38
	DataVault Pro	277	366	17	43.08

Finding: Compliance is weak everywhere (35–41%), but the most concerning pattern is by priority: compliance is **worst** for Critical-priority tickets and best for the catch-all “Unspecified” bucket — exactly backwards from what you'd want, since Critical tickets carry the tightest targets and the most at stake. By product, the 33.9%–43.1% spread is real but modest — no single product is driving the overall problem.

Query 30

Top 10 Clients by Ticket Volume

Business question: are the clients generating the most support tickets our biggest accounts, or is heavy support usage independent of contract size?

Approach: *Joins Support_Tickets to Clients on Client_ID to see each high-ticket-volume client's contract value and current account status alongside their raw ticket count.*

	Client_ID	Company_Name	Annual_Contract_Value	Account_Status	Ticket_Count
►	CLI-0041	BrightPath Labs	75000	Churned	34
	CLI-0206	GreenPulse Corp	100000	Churned	32
	CLI-0075	SwiftCode Infotech	50000	Active	30
	CLI-0223	Pinnacle Technologies	75000	Active	27
	CLI-0033	NetDynamic Group	500000	Churned	27
	CLI-0046	SwiftCode Corp	300000	Active	24
	CLI-0080	NetDynamic Labs	100000	Active	23
	CLI-0179	OmniSoft Digital	100000	Active	23
	CLI-0097	BlueOrbit Corp	75000	Active	23
	CLI-0185	BrightPath Services	200000	At Risk	23

Finding: It's a mixed picture, not cleanly “high-value” or “low-value.” 3 of the top 5 heaviest ticket-volume clients have already churned, regardless of contract size — including NetDynamic Group, one of the largest contracts in the entire top 10. This is worth flagging as a possible early-warning signal: unusually high ticket volume may predict churn risk on its own, independent of account value.

Query 31

Average Resolution Time by Issue Type

Business question: which kinds of problems take the longest to fix, and do slow-resolving issue types also leave customers unhappy?

Approach: *AVG() automatically skips blank cells, so unresolved tickets and un-surveyed customers are simply excluded from their respective averages rather than counted as zero.*

	Issue_Type	Ticket_Count	Avg_Resolution_Hours	Avg_Satisfaction
►	Access Permission Issue	334	31.8	3.68
	Data Sync Error	331	31.1	3.56
	Feature Request	309	30.3	3.56
	API Integration Failure	328	30.2	3.80
	Report Inaccuracy	341	29.2	3.65
	Dashboard Bug	325	29.1	3.56
	Onboarding Support	355	29.0	3.78
	Performance Lag	328	28.7	3.70
	Security Alert	341	28.4	3.60
	Billing Query	350	28.1	3.64
	Server Downtime	332	27.8	3.72
	Login Issue	326	27.1	3.78

Finding: Access Permission Issue is slowest overall, but doesn't have the worst satisfaction. Data Sync Error is the real standout problem: 2nd-slowest AND tied for the lowest satisfaction score — slow and poorly rated together, unlike most other issue types where slow resolution doesn't necessarily upset the customer. Login Issue is the fastest to resolve and also rates well.

Cross-Functional Queries (32-33)

Query 32

Support Escalations vs. Satisfaction and Repeat Deals

Business question: do clients with more support escalations end up less satisfied, and do they buy less from us as a result?

Approach: Summarizes each client's ticket/escalation count and deal count separately, joins both onto Clients, then buckets clients by escalation count (0 / 1-2 / 3+) and compares AVERAGES across the buckets — that comparison is the actual answer, not a raw per-client list.

	Escalation_Bucket	Client_Count	Avg_Satisfaction	Avg_Total_Deals	Avg_Deals_Won
►	0 escalations	79	3.74	7.4	1.8
	1-2 escalations	178	3.83	8.4	2.1
	3+ escalations	43	3.74	9.8	2.3

Finding: The hypothesis does **not** hold in this data. Average satisfaction is essentially flat across all three escalation buckets — no meaningful decline. Deal activity actually trends the **opposite** direction from what was expected: clients with 3+ escalations average **more** total deals than clients with zero escalations, not fewer. The likely explanation is account size, not dissatisfaction — bigger clients naturally generate more tickets and more escalations simply by having more touchpoints, which masks any real satisfaction effect. Reporting this honestly rather than forcing the expected narrative.

Query 33

Regional Health — Sales, Marketing, and Support Combined

Business question: pulling revenue, marketing spend, and support quality together, which region is actually healthiest?

Approach: Combines three separate summaries — Sales revenue by region (normalized per client, per the brief's own hint not to compare raw totals), Marketing spend by Target_Region, and Support SLA compliance by region (joined through Clients). Finance_Expenses has no Region field at all, so Finance spend is stated as genuinely excluded rather than forced with an invented mapping.

	Region	Revenue	Revenue_Per_Client	Spend	SLA_Rate
►	East	39735435	153419	21875617	39.02
	North	42879857	175020	28091081	35.38
	South	43478683	165318	17011074	39.61
	West	42286524	163268	26207724	37.22

Finding: North has the highest revenue efficiency but the **weakest** SLA compliance of any region. South is the standout: 2nd-best revenue, **best** SLA compliance, and the lowest marketing spend of any region — the healthiest region overall on every measure that could be combined. East is the weakest: the 2nd-highest marketing spend but the lowest revenue per client — the weakest return on marketing investment of the four regions.

The Bottom Line

Eight queries, read together, point to a consistent set of specific, addressable issues — not one root cause, but several:

Marketing spend and marketing efficiency are two different things

- Google Ads leads on both; Email is the most expensive channel by far despite the smallest budget. (Q26)

Brand Awareness campaigns quietly outperform

- Roughly double the ROI of the next-best campaign type, consistently across 34 campaigns. (Q27)

A real channel is going untracked

- Cold Outreach generates ₹2.07 crore in closed deals with zero recorded marketing spend. (Q28)

SLA compliance fails hardest exactly where it matters most

- Critical-priority tickets have the worst compliance of any priority level. (Q29)

High ticket volume is an early-warning signal for churn

- 3 of the top 5 highest-ticket-volume clients have already churned, regardless of contract size. (Q30)

The expected escalation-satisfaction link doesn't hold

- Satisfaction stays flat regardless of escalation count — reported honestly rather than forced. (Q32)

South is the model region; East is the one to fix

- South leads on revenue efficiency, SLA compliance, and marketing spend discipline all at once. (Q33)

Every finding above traces to a specific query, a specific number, and a specific business question.

None of it is inferred or assumed — it's what the data actually shows when the right questions are asked of it.